

Stored Procedures

- Instead of writing same SQL again & again in diff apps, u write it once; as a stored procedure & just call it by name
- Accept Parameters, which make them flexible
- improves Security

Indexes

- Performance
- Like Index of a book
- Fast lookup
- They Take storage space
- Faster Reads; Slower writes

Ansh Lamba

cloud → central storage.

- computing → Pay-As-you-Go.
Data Centers.

Azure
AWS
GCP } → cloud Providers.

cloud computing → Data Centers.



Remote Servers.

provides On-demand computing resources.

On-premise → everything is Built on ur laptop.
→ all services are " " " "
→ So, u have to Take care of all the things.

Types of cloud services

① On-Premise.

② IaaS (Infrastr)

- Azure provides just the Infrastructure.

- Eg: Virtual M/C.

- Only Phy. Hosts
Phy. N/w
Phy. Datacenters

→ will be managed by cloud.
(In this case, Azure)

③ PaaS

Eg: Azure SQL DB; Foundry
we're requesting an app.

- App/

- N/w controls (like make it accessible only to me & my team | to everyone
(Private N/w) (Public N/w))

- Identity & dir infrastructure

Shared Responsibility

(Both we & cloud will take care)

OS, Phy Host, Phy. NW, Phy. datacenter → controlled by cloud.

④ SaaS

→ u need not manage everything.

→ only, u need to manage :-

- (i) Info/ & data.
- (ii) Devices (Mobile & PC)
- (iii) Accounts & Identities

Rest all → managed by cloud.

Identity & dir infrastructure → managed by you & cloud

Eg:- Microsoft Fabric

(u need not worry abt

- i) storage
- ii) upgrades

iii) data config.)

→ u just need to handle ur acct & ur device

FAULT TOLERANCE

- Data centers.
- During disasters.
- Data Availability zones

→ Region-Pairs
If whole region is affected by disaster, still data is replicated on other region as well.

Policies

① LRS (Locally Redundant Storage) / Avail Sets
- multiple copies created within the same Data Center.

- Availability Sets (LRS also known as).

② ZRS (Zone Redundant Storage)

→ If whole DC is not working.
→ Then, go to other zone.

Tricky.

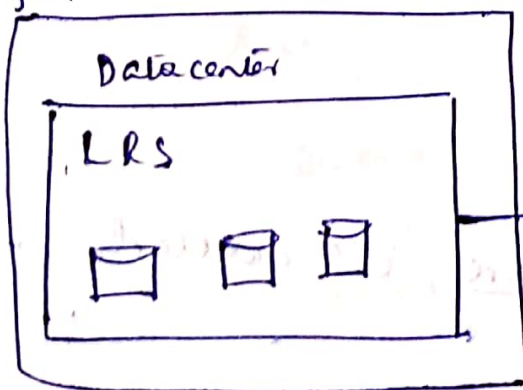
③ GRS (Geo Redundant Storage)

creates LRS within the ^{same} region, within same Datacenter.

This will be connected to diff. region, other Datacenter.

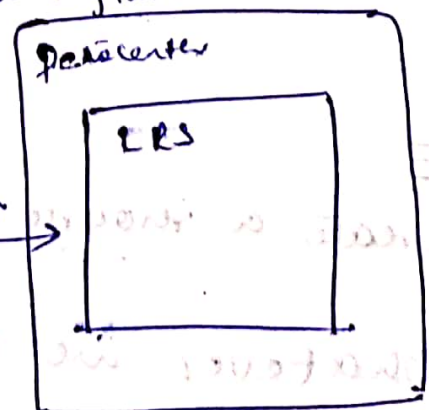
(Note: There's no zone here)

1st Region



Geo-Replication

2nd Region

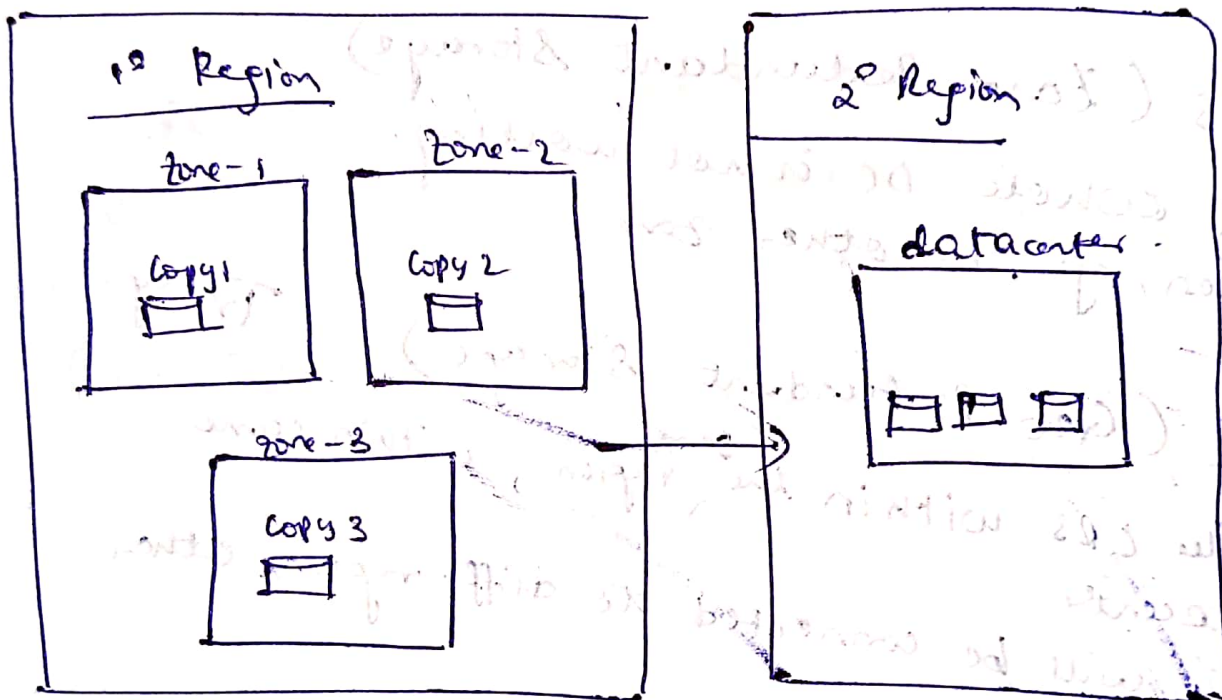


④ Geo RS (Geo Zone Redundant Storage)

- Max to 4K + Zone (Most Expensive)
↳ entire

Within same region, 3 copies within diff. zone.

Then, one data center Again 3 copies.



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≡

Create a resource +

whatever we create in Azure, it's called
Resource
↳ a service

Dashboard → collectively store all resources
→ to organise

All Resources → all of them listed here

@gvpce.ac.in } → this domain name
@outlook.in } called as tenant

Tenant .

→ Subscription :- to isolate, diff. depts
(HR, IT, Finance)

→ Resource Grp :- All resources should be placed

↓
a folder which
holds our
resources

↓
like VM; SQL

Hierarchy

Tenant



Subscription



Resource Grp



Resources

SQL DB } Resources.
Cosmos DB }

Microsoft Entra ID → renamed from

Azure Active Directory

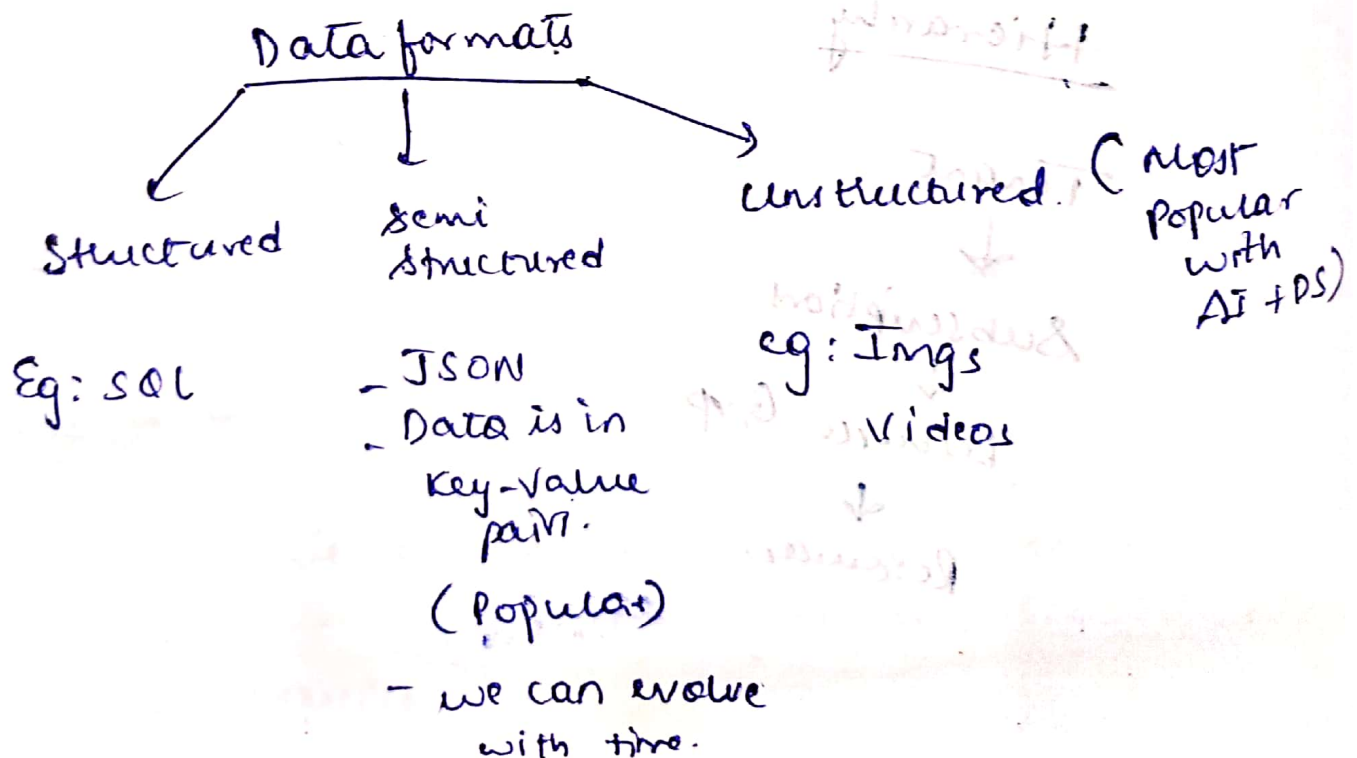
↓
we can manage
all users, apps, services,
external users
of org.

↓
This is like kind of
admin portal

Monitor → to monitor resource.

Cost Mgmt + Billing

Help + Support



Structured data in Azure:

Azure SQL DB → its on cloud.

→ ~~provid~~

→ Handles Big data.

Marketplace → from where we can get resource.

SQL VM → creates VM for you
(IaaS) → it installs Azure SQL.

SQL managed Instances
(PaaS)

SQL DB

- (i) 1st create Server.
- (ii) Then DB name.

collation

↳ set of tools we apply on data

→ Azure Data Studio

II) Semi-Structured Data (also NOSQL data)
↳ Cosmos DB. → Not only SQL
↳ called Analytical workspace → Non-Relational Data
↳ Has APIs.

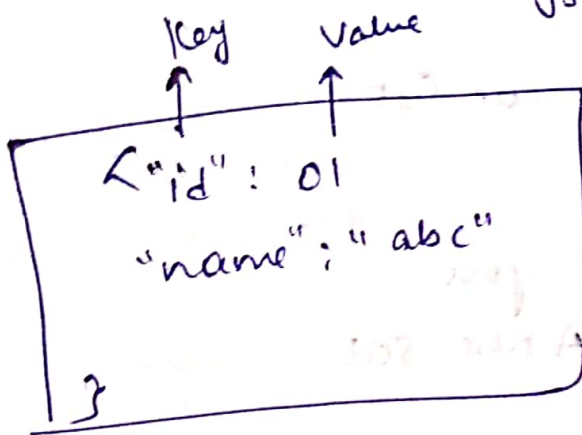
Azure Cosmos DB Recommended APIs:-

- ① Azure Cosmos DB for NOSQL.
- ② " " for MongoDB.

i) Azure Cosmos DB for NoSQL (prev - Azure Core SQL)

* → Key Value Pairs.

(, can be int
string
JSON doc



→ We can Query using SQL APIs.

eg. `select * from Table`

ii) Azure Cosmos DB for MongoDB

→ Spl type of Azure Cosmos DB for NoSQL.

* {
 "cust1": { "id": 01, "name": "abc" },
 "cust2": { "id": 02, "name": "xyz" }
}

↓
value
(it is stored as JSON)

↓
So, this becomes the Azure DB for MongoDB.

Others

(i) Azure Cosmos DB for Table

→ It is for Azure Cosmos DB for NoSQL.
→ But, ~~we~~ Here, we cannot Query the data using SQL APIs.

② Azure Cosmos DB for Apache Graph
→ when we work with Graph Databases (Nodes & Edges)

③ Azure Cosmos DB for Apache Cassandra
→ when u hv hierarchy of cols within DB.

id	customers	
	first name	last name

HTAP: Hybrid Transactional Analytical Processes.

III) Unstructured Data → Resource: Azure Storage account
data stored as:-
Blocks → when data is stored on Hard-disk.
Blobs
Files
In Azure, it's called as disks

Basis of any
Data lake

(Data lake → Built on
Top of
Blobs)

whenever we create
a azure storage acct:

By default

↓
it is blob.

But, to create Data lake,
check this box:

enable hierarchical
Namespace ☒

Access Tiers :- } Hot
Cool
Cold

- i) Hot → for I/P ; O/P data ; freq reads.
- ii) Cool → to store infrequently access data.
(for Backup)
- iii) Cold → Cheapest to store.
(Backup)
(Rarely, if u use the data)

When u create ur storage acct, u get :-

- (i) Containers → means Datalake
- (ii) File share → like central repo for files.
- (iii) Queues → msging services in Azure.
Each as source for IOT (Sensor data)
- (iv) Tables.
↳ Soln to work with semi-struc Data.
↳ continuously data will flow.

Containers → By-default (Private)

→ Note: No one shud access it ; except You & Azure

→ we can create hierarchical (nested) folders.
(folders - inside - folders).

To access Table → Go to Storage Browser

↓
from there, access Tables

Note:- u can't access Table from storage acct

Non-rel. data $\rightarrow$ we can scale our data

Once we click on Tables: we have default 3 cols:-

- (i) PartitionKey $\downarrow$ distributes uniquely.
- (ii) RowKey $\downarrow$ acts as a 1st key in each partition.
- (iii) Timestamp $\downarrow$ managed by Azure

The 1st key of whole Table (not single partition) is

PartitionKey + RowKey

$\downarrow$
also called as "Composite Key"

NoSQL $\rightarrow$ no fixed Schema

$\rightarrow$ same no of cols for all entries
Need not be maintained

$\rightarrow$ Data types:-

- Guid
- String
- Boolean
- DateTime
- Int 32
- Int 64
- Binary

$\rightarrow$ we don't have Object data type
Object data type
 $\downarrow$
available only for:-

JSON
(MongoDB)

$\downarrow$
use cosmos DB

$\leftarrow$

- it Has Partition Key
- But, no Row Key

Azure Data Services

① Azure Data Factory :- Marketplaces Data Factory

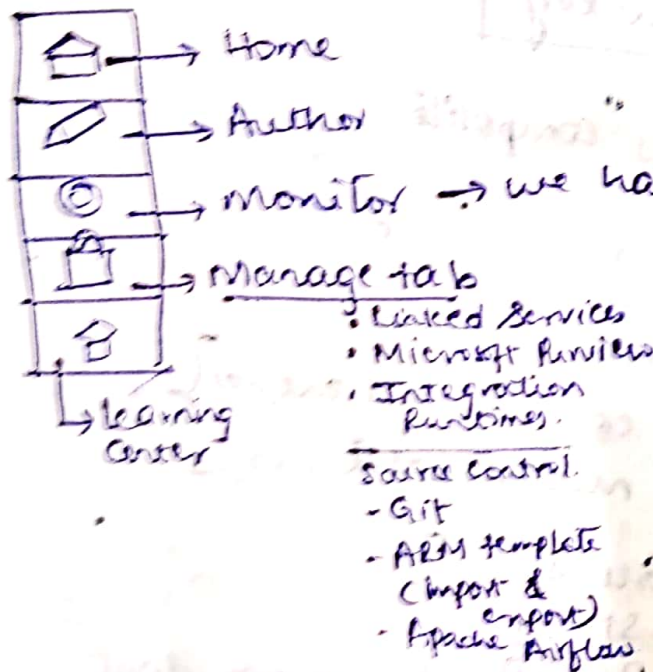
- Migrates (Transfers) data from one place to the other.

It is an ETL Tool

- Low/NO code.

- Features :-

- (i) Ingest (ii) Orchestrate (iii) Transform data.
- (iv) Configure SSIS



Runs

- Pipeline runs.
- Trigger runs.
- Change Data Capture.

Runtimes & Sessions

- Integration Runtimes
- Data flow debug.

In Azure Data Factory

linked services means

connections.

ARM → Azure Resource manager.

Azure Tab

> Pipelines

> Change Data Capture (Purview)

> Datasets

> Data flows

> Power Query

Pipelines

mapping → used to map ur data

Azure Databricks → most demanding
(if u're working with BIG DATA)

→ To process the data.
→ Manages Spark clusters.
↳ distributed computing framework

Azure Databricks full course

↓
v.v. imp.

⇒

Azure Event Hubs V.V. Imp Techno

→ To process data on real-time & Ingests data.
→ managed layer.

Streaming data

(IoT Devices,
Weather API,
Social Media)

Azure Event Hubs

→

Azure Stream Analytics

Azure Cosmos DB.

↓
Azure Blob Storage

↓
Holds data temporarily

→ Azure SQL DB

→ Power BI

Alternative
to Apache Kafka

This is consumer of data.

- Namespace $\rightarrow$ its under Event Hubs.
- We create Hubs.
- We can export events.
- Even JSON entity is treated as 1 event

Stream Analytics

- ↳ future of Azure Data Engg. when dealing with real-time data!

Azure Synapse Analytics

Azure Data factory + Spark + SQL Pool = ~~Spark~~ Synapse Analytics

SQL Datawarehouse

- ↳ renamed as : Dedicated SQL Pool.

Serverless SQL Pool $\rightarrow$ used in Data Lake

- i) Data Factory
- ii) DataBricks
- iii) Synapse

$\downarrow$
V. imp for Data Engg. role